

Exploring the effects of public transport subsidies on satisfaction and ridership

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ABSTRACT

The determinants of public transport demand are multifaceted, necessitating consideration of various factors that extend beyond the objective aspects of operation. In addition to travel time and costs, it is imperative to comprehend psychosocial factors influencing public transport usage to enhance policies geared toward fostering ridership and sustainable mobility. Empirical evidence suggests that subjective attributes, such as satisfaction and attitudes, exert a discernible influence on travel behavior and the utilization of public transport. Nevertheless, these linkages remain largely unexplored, especially within the context of large public transport networks in cities of the Global South. This paper aims to address this gap by evaluating the impact of demand-side subsidies in the form of travel vouchers on public transport ridership to understand the mediating effect of satisfaction levels according to the users' socioeconomic characteristics. These satisfaction levels are related to overall satisfaction with transport, employment and educational opportunities, and life. To achieve this, a large-scale randomized controlled experiment was conducted in Bogotá, Colombia, involving a sample of frequent public transport users ($N = 1,607$). The participants were divided into a treatment group, which received the travel vouchers loaded onto their smart travel cards for four months, and a control group. Hybrid-ordered logit models were employed to estimate the average number of weekly trips undertaken before, during, and after the intervention. The result shows an increase in ridership by the treatment group, which does not solely depend on improved affordability; but is influenced by a complex interplay of socioeconomic factors and shifts in satisfaction levels. This confirmed that travel satisfaction can contribute to increased public transport demand.

1. Introduction

The investigation into the relationships between user satisfaction and the utilization of public transport has become a crucial point in urban transport research (van Lierop et al., 2018). This is important because travel satisfaction has the potential to shape behavior and explain the desire and intention to engage in specific behaviors and attitudes toward those behaviors (De Vos et al., 2022). The links between satisfaction and transport attributes might be considered reciprocal. Traditional frameworks consider satisfaction as an outcome of transport alternatives and choices, while recent approaches suggest that the relation might go in the opposite direction, as transport satisfaction might be considered one of the main predictors of transport choices (Balbontin et al., 2023; De Vos, 2019).

This paper aims to contribute to this discussion by presenting a case study evaluating the impacts of demand-side subsidies in the form of travel vouchers on public transport ridership in Bogotá, Colombia. We

also explore the effect of these vouchers on satisfaction levels with transport, educational and job opportunities, and overall life, considering the socioeconomic characteristics of commuters. Additionally, we aim to assess the mediating effect of these subjective factors on individual public transport usage. Given that these subjective factors cannot be directly measured, we implement a latent variable approach to measure them, using a set of indicators that include dimensions that might be impacted by the vouchers. Then, we include these dimensions as latent variables, among other observed variables, as predictors in an ordered model to predict public transport travel demand.

The study utilizes a randomized controlled trial (RCT) where a treatment group received a monthly cash transfer in the form of a voucher loaded onto their personalized travel cards. For the RCT, a sample of 1607 frequent public transport users was randomly selected from the population of frequent users not receiving any other type of transport subsidy. Subsequently, half of these individuals were randomly assigned to receive the monthly voucher, while the remaining

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half constituted the control group. Baseline and closing surveys were administered before and after the intervention (voucher delivery), collecting data on revealed preferences, travel patterns, and perceptions.

Survey results were utilized to assess the impact of travel vouchers on indicators related to transport satisfaction, satisfaction with educational and job opportunities, and life satisfaction. The analysis is based on the estimation of ordered logit models, using the average number of weekly trips as the dependent variable. This approach enables us to explore the impacts and drivers of behavioral changes among public transport users from low- and medium-income countries when implementing demand-side subsidies. Moreover, we can explore the role of individual perceptions and satisfaction levels in influencing the treatment effect of a new method of delivering subsidies in one of the world's largest public transport systems. This brings fresh perspectives into research on public transport satisfaction and ridership by introducing a distinct public transport structure and cultural background. The utilization of previous behavior in a longitudinal study of the impact of a targeted subsidy represents a novel contribution to the existing literature. We also discuss the role of some intangible and satisfaction variables on public transport usage.

2. Linking satisfaction, subsidies, and public transport use

From a social perspective, public transport holds the promise of enhancing the quality of life and accessibility in urban areas (Hybel & Mulalic, 2022; Yin et al., 2021) by offering affordable transport alternatives that alleviate congestion and emissions, while also mitigating historical inequalities and transport exclusion (Chatterjee et al., 2020; Guzman, Arellana, et al., 2023). Positive user experiences characterized by reliable and comfortable services, as well as affordable fares, promote public transport use (De Vos et al., 2022). These findings align with other research associating subjective factors with travel behavior, affirming that perceived satisfaction is a determinant of public transport usage (Lai & Chen, 2011). Satisfied users contribute to high ridership levels, facilitating access to employment, education, and other opportunities, while also generating positive externalities for cities.

On one hand, objective factors related to transport alternatives, such as travel time and cost, as well as individual characteristics like income, occupation, gender, and age, typically serve as primary determinants of public transport demand in urban areas (Mwale et al., 2022). Moreover, the effects of these attributes may vary depending on the context or specific population segments. For instance, in Latin America, fares often pose a significant burden, particularly for low-income households (Estupiñan et al., 2018; Tiznado-Aitken et al., 2023). Consequently, in low- and middle-income countries within the region, public transport subsidies are considered an option to optimize the cost structure of public transport systems, ensuring good service levels with affordable fares while maintaining financial sustainability (Serebrisky et al., 2009).

On the other hand, latent or subjective factors, including perceptions, attitudes, and satisfaction, have garnered significant attention in transport research due to their social implications and their potential to elucidate the underlying determinants of travel behavior. Several studies have viewed satisfaction as an outcome of travel experience, focusing on how this experience impacts travel satisfaction levels (van Lierop et al., 2018) and subsequently affects life satisfaction (Delhey, 2004; Węziak-Białowska, 2016). Conversely, fewer studies have explored these connections in the opposite direction, examining how satisfaction influences travel behavior. There are compelling arguments suggesting that satisfaction and attitudes may have a greater influence on mode choice than vice versa (Balbontin et al., 2023; De Vos, 2019), although the utilization of these constructs as predictors of public transport remains relatively unexplored. Furthermore, the examination of subjective predictors of travel choices beyond transport satisfaction remains limited, despite the potential representation of modality styles and underlying links determining behavior.

Understanding the factors influencing public transport usage is

crucial for designing policies to leverage its potential and mitigate negative externalities associated with urban transport. There is evidence suggesting that public transport demand can be influenced by factors beyond operational and built environment characteristics, such as individual perceptions, preferences, satisfaction, as well as economic and sociological considerations (De Vos, 2019). In this line, it has been found that meeting users' expectations regarding frequency and crowdedness can increase the intention to use public transport services through the mediating effect of transport satisfaction (De Vos, 2019).

Several factors might contribute to the satisfaction of public transport users. While traditional studies primarily focused on tangible aspects such as schedules, routes, frequencies, crowding, safety, image, or travel times (Echaniz et al., 2022; Soza-Parra et al., 2019; van Lierop et al., 2018; van Lierop & El-Geneidy, 2016; Wong et al., 2023), contemporary research, particularly in the last decade, has broadened its scope to encompass more subjective attributes such as perceptions and individual preferences (De Vos et al., 2022; Fu & Juan, 2017). This shift in focus acknowledges the holistic nature of user satisfaction and its intricate interplay with diverse contextual and socioeconomic attributes. The literature on these matters is young and primarily focused on China, North America, and Europe, demonstrating robust empirical evidence for service quality and satisfaction as key factors in promoting public transport use (Carvalho dos Reis et al., 2022).

In addition to the relationship between satisfaction and the use of public transport, the analysis can be extended to other intangible attributes related to satisfaction levels. For example, Vallejo-Borda et al. (2022) observed that life satisfaction and well-being were determinants of shifts in travel behavior during the COVID-19 pandemic in South America. Employing a hybrid choice model, Balbontin et al. (2023) concluded that satisfaction with life and attitudes toward public transport plays a significant role in travel behavior, particularly in modal choice, the decision to commute, and the choice to work from home. In Bogotá, it has been identified that higher social capital stocks are associated with a greater willingness to use a new public transport service (Guzman, Cantillo-Garcia, et al., 2023). In several Italian cities, it was shown that older adults who are satisfied with their local public transport characteristics and neighborhood are more likely to have higher subjective well-being (Rossi et al., 2023).

Existing evidence suggests that intangible attributes beyond travel satisfaction such as satisfaction with life, social capital, and well-being might explain some aspects of travel behavior and modal choice. However, more empirical research is required, especially since these links can be context-dependent. In low- and medium-income countries, travel times and fares are one of the main drivers of public transport satisfaction (van Lierop et al., 2018). This is because affordability constraints are more relevant for lower-income populations, and satisfaction with travel expenses (or perceived value) plays a relevant role in overall transport satisfaction (Lai & Chen, 2011). Therefore, targeted subsidies might increase user satisfaction by making public transport more affordable and accessible (Guzman & Oviedo, 2018). Some studies have demonstrated that targeted demand-side subsidies positively impact transport and life satisfaction, especially among frequent public transport users (Mokonyama & Venter, 2018; Tavares et al., 2021). Nevertheless, there is a lack of studies evaluating the mediating effect of life and transport satisfaction on travel behavior, particularly on public transport usage. We hypothesize that travel vouchers can create virtuous circles by improving satisfaction and subsequently increasing ridership.

Examining the links between public transport subsidies and user satisfaction is imperative for policymakers and transport agencies aiming to achieve sustained financial performance and establish equitable and efficient transport systems in the Global South. The majority of studies scrutinizing the impacts of public transport subsidies have primarily concentrated on aspects such as affordability, ridership, and accessibility (Börjesson et al., 2020; Rivas et al., 2018). These studies have concluded that subsidies can increase public transport ridership by increasing budget availability or reducing travel costs. In contrast, there

is a lack of research delving into the effects of subsidies on users' satisfaction. Moreover, the behavioral determinants that drive changes in public transport usage in the context of subsidies remain unclear.

While public transport subsidies are commonly employed to encourage ridership, discerning the intricate links between subsidies, satisfaction, and ridership is complex and contingent on numerous factors. Despite evidence suggesting that travel behavior is influenced by attitudes, transport satisfaction, and life satisfaction (Balbontin et al., 2023; De Vos, 2019), there is a scarcity of research in developing cities. Additionally, these connections have not been explored within the framework of public transport subsidies. In summary, the association between satisfaction and the choice of public transport constitutes a multifaceted and dynamic subject within the domain of transport studies. This comprehensive literature review underscores the intricate interplay of various factors, encompassing service quality, accessibility, socioeconomic attributes, and methodological approaches. Despite its empirical contributions, drawing comprehending how behavior evolves based on cross-sectional design is not feasible. Addressing a gap in the literature, we present findings from a large longitudinal study of travel behavior change in a developing context.

3. Experiment design and implementation

We want to know how the travel voucher impacts public transport demand and then, transport satisfaction, satisfaction with life, and satisfaction with education and jobs, as well as explore the mediating effect of these variables on ridership and behavioral changes. To find out, we designed and implemented an RCT in Bogotá's Integrated Public Transport System (SITP by its acronym in Spanish), which moves >4 million people every day.

3.1. Case study description

The old public transport system in Bogotá fostered an excess of vehicles, low-quality service, markedly low road safety, and escalated and disparate fare structures (Ardila, 2007). This persisted for decades until the initiation of the inaugural phase of the Bus Rapid Transit (BRT) system, TransMilenio, in December 2000. This marked the commencement of the city's definitive metamorphosis in the provision of public transport services. The transformation involved a phased introduction of the Integrated Public Transportation System (SITP), comprising various components such as the BRT system, conventional buses (zonal and feeder buses utilizing diverse vehicle types), and a cable car line. The SITP introduced a centralized control mechanism for routes and vehicles, an integrated fare structure encompassing all components, and facilitated a maximum of two free transfers within a 110-min timeframe. The administration of the SITP is overseen by a public entity known as TransMilenio S.A. (TM S.A.).

Bogotá's SITP works with a flat fare. In 2021, during the our experiment, the standard fare for regular bus services was 2300 COP (~0.61 USD), while BRT services had a fare of 2500 COP (~0.67 USD). On a typical pre-pandemic day, out of the approximately 13.6 million trips undertaken in the city, 4.4 million were facilitated by public transport. Of these daily trips, 86% were undertaken by users residing in socioeconomic strata (SES) 1, 2, and 3 areas, denoting the economically disadvantaged zones. SES represents a housing classification system in Colombia, categorizing residential areas into six categories based on physical conditions and the built environment, with SES 1 denoting the least affluent and SES 6 signifying the wealthiest (Cantillo-García et al., 2019). In terms of average household income, 83% of SITP users in Bogotá belong to households earning less than USD 1070 per month. These households, typically situated in the urban periphery, predominantly rely on SITP as their primary transport mode and exhibit limited affordability levels (Guzman & Oviedo, 2018), categorizing them as captive users of this transport mode (Guzman, Beltran, Bonilla, & Gomez Cardona, 2021).

Conversely, a satisfaction study of SITP users in Bogotá spanning from 2017 to 2022 reveals that the overall user satisfaction level, encompassing both regular buses and the BRT component, has remained consistently at around 68% on a scale ranging from 0 to 1 (see Fig. 1). While there is a discernible variance in satisfaction levels between the regular bus and BRT components, with the bus service being perceived as inferior, a recent trend indicates a narrowing of the satisfaction gap between the two components.

The satisfaction index primarily measures service quality in operational terms, encompassing schedules, routes, travel and waiting times, infrastructure conditions, signaling, and personnel. Nevertheless, these satisfaction metrics do not incorporate crucial elements such as fare costs or access to opportunities. Notwithstanding, data released by TM S. A. posits that users would perceive SITP more favorably if the city offered more employment opportunities (93%) and if the fares were more affordable (92% of users).

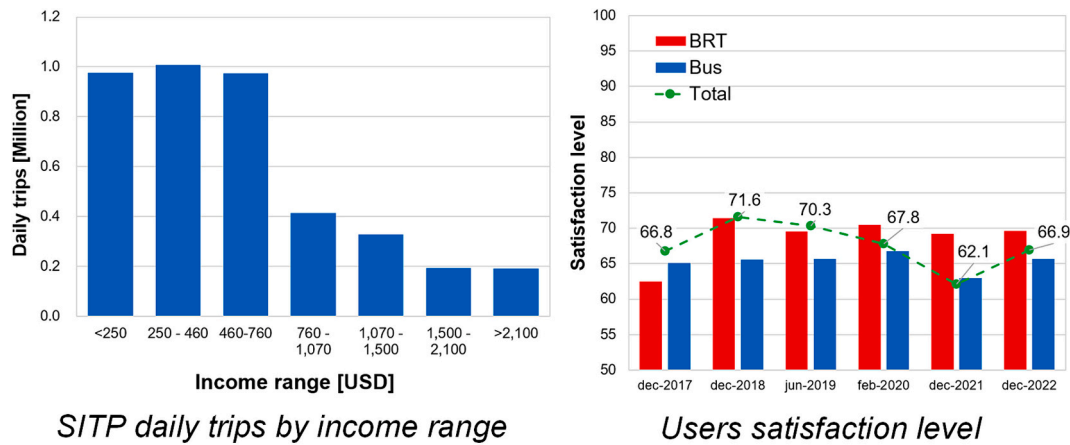
3.2. Experimental design

We used a large-scale RCT of Bogotá's public transport system's frequent users without any transport subsidy. From the universe of Bogotá's public transport system's frequent users (around 176,000 users), we randomly selected a sample of 1607 individuals. In this case, half of the sample was randomly assigned to a treatment group ($n = 801$) and the other half to a control group ($n = 806$). The sampling process was done using a probabilistic approach assuming a uniform distribution, guaranteeing similarity between the two groups. For four consecutive months starting in May 2021, the users in the treatment group received a monthly cash transfer into their public transport smart card (the travel voucher), meaning that everyone in this group received four transfers, while the control group did not receive any subsidy. In turn, we divided the participants of the treatment group into two categories: Treatment A, 402 participants who received a monthly travel voucher of 28,000 Colombian pesos (COP) (~7.5 USD). In treatment B, 399 participants received a voucher for 21,000 COP (~5.6 USD).

Before the intervention started (travel voucher delivery), participants in the experiment were contacted to secure their consent, provide an explanation of the trial, and validate their profiles and eligibility. Participants in this study agreed to the following:

- The administration of a household survey before the initial voucher distribution to capture baseline information for characterizing sociodemographics, household attributes, travel patterns, and a set of attitudinal indicators (conducted during March and April 2021).
- The implementation of a closing survey after the RCT was completed, collecting follow-up information similar to that obtained at the baseline. This phase occurred from October to November 2021.
- Daily reporting on public transport usage throughout the intervention period (from May to August 2021). This was carried out for all participants through an analysis of their travel card records. This information is available for one month before and after the intervention for each participant.
- Participants in the treatment groups received a monthly cash transfer into their personalized travel card from June to September 2021.
- As an incentive, participants from the control group received a gift card (COP 30,000–7.8 USD) to be redeemed at various supermarkets at the end of the trial.

The experiment was divided into three phases: baseline (T0), intervention (T1), and closure (T2). At the baseline, the household survey was administered to all participants ($N = 1607$). The questionnaire comprised four sections, starting with an introduction by the surveyor and an overview of the study. The second section inquired about travel patterns, main trip purposes and activities, transport modes used, travel times, average transport expenditure, and indicators of travel satisfaction. The third section gathered participants' perceptions regarding a set



Source: Household travel survey 2019 and TM S.A.

Fig. 1. Daily trips by SITP and satisfaction level.

of indicators related to the quality of life, work, education, accessibility to opportunities, basic needs, and free time use. Finally, the last survey section collected sociodemographic information, including household income, residence location, time living in the current house, household size, gender, age, marital status, and occupation.

The intervention phase commenced in late May 2021, after the collection of the baseline survey, with the issuance of travel vouchers to the participants' personalized travel cards. At the end of the intervention, a closing survey, mirroring the baseline, was conducted as is shown in Fig. 2.

The information obtained at this stage was compared with the baseline to discern the impact of the trial on travel patterns, satisfaction levels, and attitudes. The final sample for the closing survey comprised 1414 individuals. Detailed daily tracking of each participant's travel patterns (date, time, and station) on SITP was conducted using their travel card records one month before the intervention, during the four months of the intervention, and one month after the intervention concluded.

3.3. Data summary

The proposed analysis relies on baseline and closing surveys, as well as the number of trips linked to participants' travel cards. To evaluate the impact of the travel vouchers, the final sample from the closing survey ($n = 1414$) is utilized, providing panel information before and after the experiment. With an attrition rate of 18.6%, deemed low in comparison to existing literature (Abebe et al., 2021; Brough et al., 2022).

Table 1 shows a summary of statistics for participant groups, with p -values testing differences between attributes, as well as p -values testing for differences between groups by attributes. In the case of categorical

attributes, the p -value corresponds to a χ^2 test, while for numerical variables to a t -test. No statistical differences between treatment and control groups are observed, affirming their similarity.

The composition of the control and treatment groups is similar, validating the random sampling process. Approximately 40% report monthly household incomes below one monthly minimum wage (COP 908,500–240 USD). Socioeconomic strata (SES) further contextualize the sample, considering the main economic activities of occupied and non-occupied participants. Note that the main economic activity referred to in the typical transport modes used can be working or studying for occupied participants, and personal business or leisure for non-occupied. The distribution of the sample is representative of public transport users in the city according to other sources such as the last official Household Mobility Survey collected in 2019.

Table 2 displays results for attitudinal indicators used to construct the latent variables, comparing means and standard deviations between baseline (T0) and closing surveys (T2). In all cases, the indicators were coded so that higher scores indicate higher satisfaction or better perception. The groups were compared using a two-sample t -test to test the hypothesis of differences in means between them. Preliminary findings suggest increased satisfaction levels for the main transport mode and transport expenditure in treatment groups, hinting at a positive impact of the transport voucher.

These indicators will be employed to measure latent constructs explaining participant perceptions in subsequent sections. This contributes to unraveling underlying factors shaping travel behavior and public transport usage in the context of the travel voucher implementation. The incorporation of latent variables aims to assess the hypothesis that travel behavior is influenced by attitudes and perceptions (Fu & Juan, 2017).

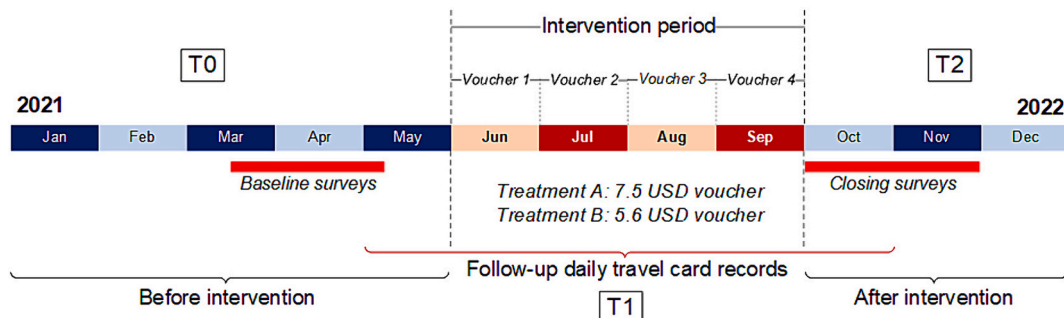


Fig. 2. RCT timeline.

Table 1
Sample description.

Attribute		Control	Treatment A	Treatment B	p-value
Age (years)	18–25	7%	8%	7%	0.942
	25–30	6%	8%	8%	
	30–40	24%	24%	20%	
	40–50	27%	26%	30%	0.735
	50–60	28%	26%	28%	
	> 60	8%	8%	7%	
Gender: Female		75%	75%	72%	0.942
Occupation: Working or studying		39%	51%	40%	0.376
Education level	Primary	10%	9%	11%	0.632
	Secondary	28%	24%	25%	
	Undergraduate	30%	20%	24%	
	Postgraduate	3%	1%	2%	
	None / No response	30%	43%	37%	
Socioeconomic strata (SES)	SES 1	10%	8%	8%	0.681
	SES 2	48%	37%	50%	
	SES 3	12%	11%	9%	
	SES 4	1%	1%	1%	
	Not applicable / No response	28%	47%	36%	
Monthly household income	Less than COP 908,526 (240 USD)	37%	39%	39%	0.626
	COP 908,526 (240 USD) to COP 1,500,000 (390 USD)	31%	32%	29%	
	COP 1,500,000 (390 USD) to COP 2,000,000 (520 USD)	15%	13%	15%	
	More than COP 2,000,000 (520 USD)	17%	15%	16%	
	Public transport	83%	86%	85%	
Typical transport mode used for the main economic activity	Bicycle	2%	1%	2%	0.641
	Walking	4%	2%	3%	
	Car	6%	7%	5%	
	Motorcycle	1%	1%	2%	
	Other	4%	3%	3%	
Average household size	Mean (st. dev.)	3.6 (1.4)	3.6 (1.6)	3.7 (1.5)	0.746
	No response	28%	43%	36%	
Average number of weekly public transport trips	Before intervention (T0)	5.3	5.6	5.4	0.541
	Intervention (T1)	6.3	7.9	7.8	<0.01***
	After intervention (T2)	6.8	7.4	6.8	
Final sample size		700	357	357	

Significance level differences between groups: 95% (*), 99% (**), 99.9% (***).

Table 2
Summary of attitudinal indicators.

Indicator	Control		Treatment A		Treatment B	
	Baseline T0	Closure T2	Baseline T0	Closure T2	Baseline T0	Closure T2
Satisfaction with the mode used to travel to the main activity regularly (1: Totally dissatisfied - 5: Totally satisfied)	3.52 (1.12)	3.41 (1.13)	3.55 (0.99)	3.63 (1.09)	3.47 (1.08)	3.54 (1.09)
Satisfaction with transport expenditure (1: Totally dissatisfied - 10: Totally satisfied)	5.46 (2.38)	5.80*** (2.35)	5.30 (2.22)	6.18*** (2.21)	5.48 (2.37)	5.92*** (2.33)
Satisfaction with your current quality of life (1: Totally dissatisfied - 10: Totally satisfied)	6.31 (1.91)	6.79*** (2.09)	6.45 (1.76)	7.15*** (1.99)	6.48 (1.72)	6.89*** (2.09)
Satisfaction with your family life (1: Totally dissatisfied - 5: Totally satisfied)	8.79 (1.58)	8.61*** (1.64)	8.78 (1.68)	8.76 (1.47)	8.71 (1.68)	8.65 (1.67)
Satisfaction with other expenses (1: Totally dissatisfied - 5: Totally satisfied)	5.85 (2.22)	6.18*** (2.09)	5.96 (2.2)	6.37*** (2.1)	5.97 (2.11)	6.11 (2.15)
I am optimistic regarding my future (1: Totally agree - 5: Totally disagree)	4.11 (0.85)	3.93*** (1.21)	4.13 (0.86)	3.87*** (1.31)	4.11 (0.87)	4.07 (1.13)
You can trust most people (1: Totally agree - 10: Totally disagree)	3.98 (2.47)	4.15 (2.6)	4.02 (2.58)	4.13 (2.60)	3.94 (2.46)	4.16 (2.53)
Satisfaction with your education (1: Totally dissatisfied - 10: Totally satisfied)	7.26 (2.31)	7.24 (2.32)	7.2 (2.38)	7.35 (2.37)	7.28 (2.25)	7.34 (2.14)
Satisfaction with your current job (1: Totally dissatisfied - 10: Totally satisfied)	7.51 (2.48)	7.62 (2.28)	7.81 (2.21)	7.7 (2.19)	7.82 (2.13)	7.64 (2.17)

Mean (standard deviation) Significance level differences between times by group: 95% (*), 99% (**), 99.9% (***).

4. Modeling approach

This paper aims to evaluate the impact of travel vouchers on public transport ridership and to understand the mediating effect of satisfaction levels according to the socioeconomic attributes of commuters. To achieve this, we employ a hybrid ordered logit model with the average

number of daily trips made by public transport as the dependent variable. This variable is commonly used to assess and forecast travel demand (Bwambale et al., 2019). Additionally, we include observed attributes as predictors, such as travel vouchers, and three latent variables related to satisfaction with transport, opportunities, and life. This methodological approach enables us to assess the relative effect of these

variables on ridership and the travel behavior of participants. The modeling results are valuable for predicting ridership in the context of public transport subsidies, as well as understanding the underlying factors that mediate satisfaction levels.

In this case, the model is estimated using data from three distinct periods: before the voucher delivery (T0), during the intervention (T1), and when the intervention ended (T2). Initially, the baseline (T0) encompasses trips undertaken a month before the intervention. Subsequently, during the intervention phase (T1), we observe the trips conducted over the four-month period during which vouchers are distributed to the treatment group. Finally, the closure phase (T2) pertains to the average number of trips made one month following the conclusion of the experiment (see Fig. 2). This methodological framework enables us to assess the determinants influencing travel behavior, quantified as the average number of weekly trips during each specified period.

Fig. 3 illustrates the structure of the proposed model to evaluate the mediating effect of satisfaction (latent variables) on ridership. It is noteworthy that the error terms are omitted to enhance the clarity of the visual representation of the relationships between variables. As discussed earlier, the quantity of weekly trips is a function of the observable characteristics of participants, the travel voucher, and latent variables. The explanatory variables and indicators are derived from the collected panel survey. Baseline information from the indicators was utilized to model the latent variables and trips undertaken at T0 and T1 for the control group, and at T0 for the treatment group. Similarly, data from the closing survey was employed to model trips at T0 for the control group, and at T1 and T2 for the treatment groups. It is assumed that the closing survey captures changes in perceptions experienced by participants in the treatment group due to the intervention.

Following the model structure depicted in Fig. 3, transport satisfaction is linked to perceptions of the predominant transport mode and transport expenses. Approximately 85% of the sample identifies public transport as the primary mode, making this variable associated with satisfaction regarding the public transport service. Although transport satisfaction can be linked to various attributes such as comfort, safety, waiting time, and transfers, no significant impact on these attributes was anticipated in this instance, as the experiment focused solely on the expense component with no changes to the supply or level of service.

Moreover, life satisfaction is correlated with quality of life, with the definition of this latent construct based on the conceptual framework established by Campbell et al. (1976). They stated that quality of life can

be characterized by directly exploring individuals' perception of life in general, and assert that neighborhood satisfaction and transport characteristics are key determinants of life satisfaction. While some studies have associated life satisfaction with travel behavior, most literature links the effects of travel patterns on quality of life (De Vos, 2019). Fewer studies assess the reverse relationship—namely, the influence of life satisfaction on travel behavior. Similarly, satisfaction with opportunities was defined as a metric to assess access to educational and job opportunities in the city, with the consistency and reliability of latent variables evaluated through confirmatory factor analysis (CFA).

The assessment of transport satisfaction and perceptions is intended to test certain theories positing it as a key factor influencing travel behavior (De Vos, 2019; De Vos et al., 2022; Fu & Juan, 2017). Furthermore, by incorporating a dummy variable for the treatment group at T2, we can evaluate the impact of the travel voucher on these latent variables and ascertain if the intervention affects participant satisfaction.

Regarding the model specification, given that the response variable, that is, the average number of weekly trips made by the participants (y_i^*) is a continuous variable, we first transform it into an ordered variable by grouping the values into six categories using 0, 5, 10, 15, and 20 trips as cutoffs. Then, the transformed variable y_i for participant i can be modeled using the ordered logit formulation described by Eq. (1) and Eq. (2).

$$y_i^* = \sum \beta_x X_i + \zeta_i \quad (1)$$

$$y_i = \begin{cases} 0 & \text{if } y_i^* \leq \tau_1 \\ 1 - 5 & \text{if } \tau_1 < y_i^* \leq \tau_2 \\ 6 - 10 & \text{if } \tau_2 < y_i^* \leq \tau_3 \\ 11 - 15 & \text{if } \tau_3 < y_i^* \leq \tau_4 \\ 16 - 20 & \text{if } \tau_4 < y_i^* \leq \tau_5 \\ 20 + & \text{if } \tau_5 < y_i^* \end{cases} \quad (2)$$

Where X_i is a set of explanatory variables, β_x the set of corresponding parameters to estimate, ζ_i is a random error component that follows the Gumbel distribution, and τ_k are thresholds to estimate. Note that y_i^* from Eq. (1), can be seen as a latent trip-making propensity for individual i , which is then converted into ordered trips using the thresholds. Under these assumptions, the probability of observing category k for the

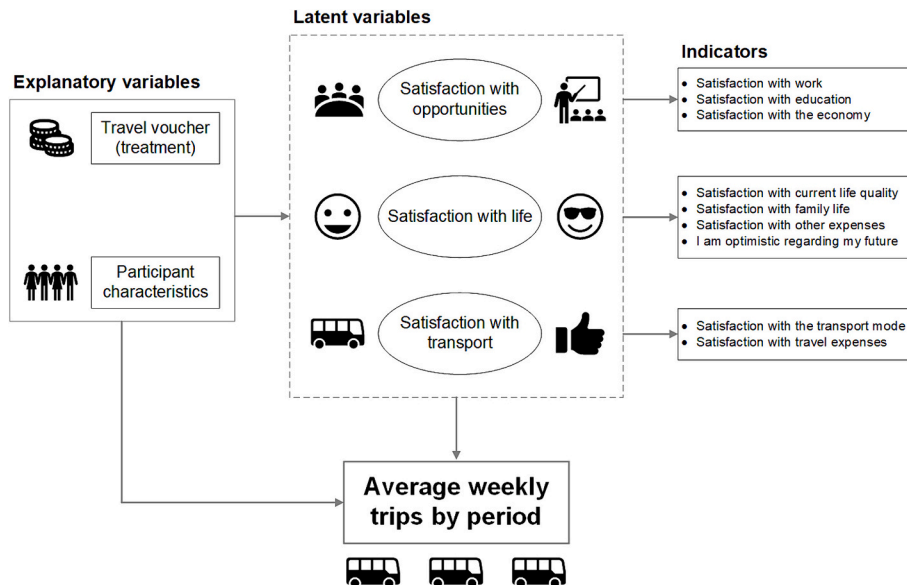


Fig. 3. The proposed model structure.

observed variable y_i is given by Eq. (3):

$$P(y_i = k) = \frac{e^{\tau_k - y_i^*}}{1 + e^{\tau_k - y_i^*}} - \frac{e^{\tau_{k-1} - y_i^*}}{1 + e^{\tau_{k-1} - y_i^*}} \quad (3)$$

To analyze the impact of explanatory variables (X_i), we examine the influence of sociodemographic characteristics of participants, the availability of travel vouchers, and latent variables representing subjective attitudes and underlying perceptions that potentially influence travel behavior, yet are not directly observable. The data for these explanatory variables are derived from the collected surveys, while the latent constructs are measured through attitudinal indicators from Table 2. We assess three latent variables: transport satisfaction, life satisfaction, and satisfaction with educational and employment opportunities.

The structural Eq. (4) shows the latent variables η_i which are specified as a function of a set of observed variables S_i affected by the parameters α_i , plus a normal error term ϑ_i . The incorporation of S_i allows for capturing heterogeneity in the latent variables by observed attributes (Bollen, 1989).

$$\eta_i = \sum \alpha_i S_i + \vartheta_i \quad (4)$$

Furthermore, the latent variables explain the set of attitudinal indicators C_i , which are treated as continuous variables, through a normal probability density function. This implies that the probability of observing the specific values of the indicator is given by Eq. (5), also called measurement equations, where μ is the mean of the indicator, σ the standard deviation, and ϕ the Standard Normal density function (Hess & de Palma, 2019).

$$P(C_i) = \frac{\phi\left(\frac{C_i - \eta_i - \mu}{\sigma}\right)}{\sigma} \quad (5)$$

Then, the probability of observing the vector of categories y_i is given by Eq. (6), where φ is the density function of the latent variables. The set of parameters α_i can be estimated by maximizing the loglikelihood of the set of choices made by the individuals considering the probabilities from Eq. (5) (Train, 2009).

$$P(y_i) = \int \sum_{k=1}^K P(y_i = k | y_i^*) \phi(C_i | \eta_i) \varphi(\eta_i | S_i, \alpha_i, \vartheta_i) d\eta_i \quad (6)$$

Table 3

Modeling results. Parameter estimated for the structural equations of latent variables.

Variable	Parameter	Coef.	t-test	p-value
Life satisfaction	Income ≤ 1 monthly minimum wage	0.02	2.77	0.006**
	Female	-0.03	-4.14	<0.001***
	Working or studying	0.05	8.33	<0.001***
	Treatment after intervention	0.04	6.69	<0.001***
Satisfaction with transport	Income ≤ 1 monthly minimum wage	0.28	15.43	<0.001***
	Female	-0.11	-6.88	<0.001***
	Working or studying	0.17	12.14	<0.001***
	Treatment after intervention	0.17	12.42	<0.001***
Satisfaction with opportunities	Income ≤ 1 monthly minimum wage	0.16	11.97	<0.001***
	Female	-0.03	-2.69	0.007**
	Working or studying	0.11	10.87	<0.001***
	Treatment after intervention	0.08	7.50	<0.001***

Significance levels: 95% (*), 99% (**), 99.9% (***).

5. Results

Table 3 presents the parameter estimates (α_i) derived from the structural equations (Eq. 4) defining the three latent variables. These estimates serve the purpose of characterizing the sample and analyzing satisfaction levels, while also enabling an examination of variations in perceptions based on participants' socioeconomic attributes. Furthermore, they facilitate the interpretation of the outcomes of the ordered logit models. Through these estimates, we can discern the socioeconomic characteristics that exhibit a higher likelihood of influencing satisfaction. Regarding the influence of travel vouchers on satisfaction levels, we introduced a dummy variable for treatment after the intervention (T2). This parameter tests for statistical differences in this group post-intervention. While differences between Treatment A and Treatment B were tested, no statistical differences were found, suggesting a similar effect for both groups. The variable *Treatment after intervention* was intentionally designed to measure the impact of the travel voucher on satisfaction levels within each latent construct. It assumes a value of 1 if the participant is part of the treatment group during period T2.

Results in Table 3 confirm that travel vouchers positively impact satisfaction levels (positive α values), supporting the hypothesis that this policy enhances perceptions of public transport services. Given that the majority of the sample identifies SITP as their primary transport mode, it can be argued that transport satisfaction extends to public transport satisfaction. In other words, travel vouchers enhance individuals' perceptions of public transport affordability, elevating satisfaction with transport expenditure and payment capacity. This is particularly significant for a system with various areas for improvement. It should not be overlooked that the intervention does not affect supply-related factors (service level, comfort, safety, or reliability) influencing satisfaction.

Detailed analysis of the results in Table 3 reveals that individuals with incomes equal to or lower than one monthly minimum wage, who are employed or studying, which implies a high probability of having an informal job (no regular income), exhibit higher satisfaction with transport (+0.28). This lower-income population segment relies more heavily on public transport and expresses greater satisfaction with the services it provides. Additionally, employed participants report higher satisfaction with life and opportunities compared to those not working or studying (+0.05 and +0.11, respectively).

Female participants exhibit lower satisfaction across all evaluated latent constructs, highlighting gender disparities in access to employment and education in the city. According to the results, males are around 47% more satisfied with transport compared to female participants. The reduced satisfaction of women with transport could be linked to a preference for alternative modes due to safety and harassment concerns (Guzman, Arellana, Cantillo-García, & de Ortúzar, 2021; Quinones, 2020). Women also appear less satisfied with life (-0.03), which reflects gender inequalities in low-income contexts (Guzman, Arellana, et al., 2023). Public transport subsidies have the potential to mitigate partially these gender gaps.

Occupied individuals report greater satisfaction with their standard of living compared to those not engaged in work or study. This may be attributed to higher income, budget, and greater satisfaction with general expenses. Notably, participants with lower incomes report higher satisfaction levels in all three latent variables, aligning with global research on happiness, well-being, and satisfaction. It has been shown that, in general, people want the same things: a decent standard of living, a good family and social life, a good neighborhood, a job, and good health (Borooah, 2006). Additionally, in developing contexts with significant inequalities where people face various challenges, they seek to enjoy life, focus on non-material values, and strive for harmony, which could contribute to their overall happiness (Eternod, 2021).

Furthermore, travel vouchers have the potential to influence perceptions beyond transport, such as life satisfaction and opportunities. Existing evidence suggests a correlation between improved transport

conditions and a higher quality of life in developed contexts (Hybel & Mulalic, 2022; Węziak-Białowolska, 2016). Moreover, from an objective standpoint, travel vouchers free up a budget for accessing other goods and services, generating surplus and improving overall welfare. However, our analysis focuses on the short-term effects of transport vouchers, emphasizing the need for further evaluation of long-term impacts on these aspects.

Once the impact of the travel voucher on the latent variables is elucidated, Table 4 presents the outcomes of the logit-ordered model derived from Eq. (1). Given the assumption of a linear utility, the coefficients of the estimated parameters indicate the marginal utility corresponding to the attributes under consideration. Primarily, the treatment group is more inclined to increase public transport ridership, as anticipated due to the provision of the travel voucher (+0.41 and +0.31 during and post-intervention, respectively). This voucher increases the travel budget, thereby improving affordability and accessibility for participants by alleviating costs associated with SITP usage. Importantly, a notable lag effect of the voucher persists even after the experiment concludes (*Treatment: After the intervention*: +0.31).

In terms of sociodemographic attributes, individuals with lower incomes engage in more public transport trips (+0.28 vs. +0.21). Similarly, occupied participants also manifest an increased frequency of trips (+0.14). Concerning age, individuals aged 18 to 25 undertake fewer public transport trips compared to their older counterparts (+0.33 vs. other age groups). The propensity to utilize public transport increases with age, peaking at 60 years, beyond which the number of weekly trips declines.

The parameter associated with female participants implies a reduced propensity for this group to undertake public transport trips (−0.05). This finding is consistent with prior research indicating lower preferences among women for public transport alternatives (Guzman, Arellana, Cantillo-García, & de Ortúzar, 2021) and less favorable perception of travel satisfaction than men in Bogotá (Guzman, Arellana, et al., 2023). Although this parameter lacks statistical significance (t -test = 0.603), we consider it important to mention it and take it into account for future gender policy design, despite it may imply that distinctions between males and females are not important in this context. This might be a consequence of the similar characteristics in terms of income and occupation. However, it is important to mention the mediating effect of the latent variables evaluated. In all cases, female participants showed lower satisfaction in the three aspects evaluated (see Table 3). This

means that the lower willingness of female public transport users to use the system could be a consequence of lower satisfaction with transport, life, and opportunities, rather than a direct effect of the operational characteristics of the system. This implies that reducing this gap is necessary to improve the satisfaction of women and their quality of life, which can be done partly through targeted subsidies and complementary policies that enhance the safety, comfort, and improved level of service of public transport.

Concerning latent variables, all three constructs of satisfaction exert a positive influence on the willingness to utilize public transport. This signifies that higher levels of satisfaction increase the likelihood of using the public transport system. While the marginal contribution of life satisfaction is the most pronounced among the latent variables, it is significant solely at the 90% confidence interval. Consequently, its impact is comparatively less robust in comparison to satisfaction with transport, which achieves a significance level of 95% interval. In the case of satisfaction with opportunities, the parameter's significance is low (p -value = 0.213), implying that the actual effect of the variable is nearly null. Consequently, it can be deduced that satisfaction with opportunities does not strongly determine the demand for public transport.

The influence of the attribute satisfaction with transport (+0.22) is particularly noteworthy, as we have identified that the travel voucher directly enhances transport satisfaction levels, particularly due to the transport expenditure indicator. The vouchers elevate transport satisfaction, resulting in increased ridership. By comparing female and male participants included in the treatment groups, we also found that the effect of the voucher on transport satisfaction is higher for females. On average, after the RCT, the change in transport satisfaction compared to the baseline is 20% higher for females. Thus, it can be inferred that the higher use of SITP by the treatment groups during the experiment is not solely attributable to a higher budget but is also a consequence of an increase in satisfaction with the service. In other words, the effects of the vouchers extend beyond affordability improvements to include satisfaction where a differential effect by gender is evidenced, in the benefit of female users. A parallel observation can be made concerning the mediating effect of life satisfaction (+0.58).

In summary, travel vouchers heighten the likelihood of utilizing public transport not only due to enhanced affordability but also because there is an impact on travel behavior mediated by satisfaction with transport and life. This scenario also explains part of the delayed effect observed after the experiment concluded, given that the treatment group exhibited an increased number of trips the month following the trial's conclusion. These findings confirm the advantages of improving transport satisfaction for frequent users, particularly in terms of travel expenditure satisfaction. Although not within the scope of this study, it is noteworthy that strategies aimed at improving the overall quality of public transport services have also been found to positively influence both captive and non-captive users (Mokonyama & Venter, 2018). Furthermore, it is confirmed that travel satisfaction can contribute to increased public transport ridership. While our study corroborates this hypothesis in the short term, existing reviews suggest that the impact can also manifest in the long term.

6. Discussion and policy implications

Mobility in cities of the Global South, home to the majority of the world's population, confronts significant challenges. Issues such as pollution, congestion, and road fatalities are intricately linked to the rapid surge in individual motorization, notably the proliferation of motorcycles. The enhancement of affordable and reliable public transport not only improves mobility but also fosters social cohesion by encouraging interaction and community engagement (Guzman, Cantillo-García, et al., 2023; Miciukiewicz & Vigar, 2012). Moreover, the reduction in the perceived cost of public transport can particularly benefit vulnerable segments of the population, such as those not engaged in work or education, thereby improving their accessibility to

Table 4
Modeling results. Parameter estimated for the ordered logit model.

Parameter	Coef.	t-test	p-value
Satisfaction with transport	0.22	2.06	0.040*
Satisfaction with life	0.58	2.04	0.041*
Satisfaction with job and educational opportunities	0.23	1.25	0.213
Treatment: During the intervention	0.41	4.60	<0.001***
Treatment: After the intervention	0.31	3.26	0.001**
Income <1 monthly minimum wage	0.28	3.73	<0.001***
Income 1–2 monthly minimum wage	0.21	2.72	0.006**
Female	−0.05	−0.52	0.603
Working or studying	0.14	2.21	0.027*
Age 25–30	0.33	2.16	0.031*
Age 30–40	0.72	5.92	<0.001***
Age 40–50	0.86	7.08	<0.001***
Age 50–60	0.85	6.90	<0.001***
Age 60–70	0.58	3.79	<0.001***
τ_1 : Threshold 0 1–5	0.98	4.03	<0.001***
τ_2 : Threshold 1–5 6–10	2.09	8.58	<0.001***
τ_3 : Threshold 6–10 11–15	3.09	12.51	<0.001***
τ_4 : Threshold 11–15 16–20	4.30	17.02	<0.001***
τ_5 : Threshold 16–20 >20	5.54	20.97	<0.001***
Loglikelihood whole model		−79,747.09	
Loglikelihood choice (ordered logit)		−6047.28	

Significance levels: 95% (*), 99% (**), 99.9% (***).

opportunities within the city. In addition, it can mitigate gender gaps associated with differences in satisfaction levels as well as accessibility, considering that our results suggest that the lower willingness of female participants to use public transport could be a consequence of lower satisfaction with transport and life in general.

In Latin America, there exist high elasticities to fare and frequency, and public transport is generally categorized as an inferior good (Batarce & Galilea, 2018; Crôtte et al., 2009; Guzman et al., 2020; Toro-González et al., 2020). For instance in Colombia, between 2019 and 2023, the number of passengers utilizing public transport decreased by 7.1%, while the fleet in service increased by 12.7%. This ongoing trend has led to the discontinuation of public transport services in various cities. This scenario underscores a price-sensitive travel demand environment, where positive responses are observed with improvements in frequency, indicative of better service (Crôtte et al., 2009; Toro-González et al., 2020).

Notably, public transport satisfaction and ridership are not solely contingent on affordability; therefore, subsidy programs must be complemented by initiatives to enhance other service attributes influencing travel demand, including reduced travel times, improved comfort, safety, reliability, and overall service quality. The combination of subsidies and service enhancements may encourage a heightened awareness of public transport usage and potentially stimulate a shift from unsustainable transport modes. Consequently, a comprehensive approach involving public transport subsidies and increased restrictions on the use of cars/motorcycles may be imperative, as indicated by several studies. However, the application of these policies requires identifying existent inequalities regarding accessibility and satisfaction to target the initiatives to vulnerable or disadvantaged population segments.

Nevertheless, beyond considerations of affordability and operational efficiency, it is crucial to ascertain the determinants of satisfaction to establish appealing public transport systems. As demonstrated in our case study, satisfaction with opportunities and overall life satisfaction are important elements incentivizing public transport utilization. While many of these perceptions extend beyond the transport sector, our findings underscore the significance of interdisciplinary collaboration. Collaborative efforts spanning health, education, employment, recreation, and transport sectors could be very important for enhancing citizens' quality of life and concurrently promoting long-term urban sustainability. Developed countries have undertaken extensive research in this domain, and this study elucidates key elements that can inform decision-makers in developing cities aiming to enhance service quality and ensure sustained financial performance. New travel behavior paradigms should take into account the fact that attitudes, measured through the concept of satisfaction, might allow us to understand modality styles, as well as identify the determinants of transport choices beyond operational attributes, capturing the underlying psychological aspects of travel behavior.

The outcomes and recommendations of this study should be regarded as an opportunity to encourage sustainable transport behavior. By leveraging existing modal share and a substantial public transport market share, and by identifying the underlying factors influencing public transport ridership, Global South countries can devise market-oriented strategies. Investment in satisfaction and service quality, along with addressing other satisfaction-related issues, has the potential to influence public transport use, establishing a virtuous cycle.

7. Conclusions

We investigated the impacts of a novel form of transport subsidy, specifically in the form of vouchers, on public transport ridership and satisfaction in a developing context. Moreover, we evaluated the mediating effect of three satisfaction levels, particularly on the decision to change public transport usage in an experimental context. This research involved a field RCT in Bogotá that induced a temporary alteration in behavior. The intervention was meticulously designed and

implemented to assess the influence of demand-side public transport subsidies on system ridership and satisfaction. To increase participants' satisfaction with their commutes, perceptions, and attitudes, measurements were taken both before and after the treatment. Our findings substantiate that the implementation of travel vouchers leads to a significant increase in system ridership, persisting for a few weeks even after the experiment's conclusion.

Female participants and individuals who are not employed or studying exhibit a reduced likelihood of utilizing public transport extensively. This result highlights historical gender gaps in the context of low- and medium-income countries concerning accessibility. In particular, women in many of these countries engage more heavily in non-paid caring activities, so the effect of non-occupied participants is mainly driven by females. Furthermore, we have seen that the voucher positively impacts women more than men in terms of satisfaction with transport. Furthermore, the lowest-income participants (the poorest among the poor) benefited the most. These findings hold potential relevance for targeted subsidy programs aimed at addressing accessibility inequalities based on gender, income, or occupation, thereby enhancing access to opportunities for the unemployed and those involved in care activities. Our observations indicate that vouchers contribute to heightened satisfaction with the primary mode of transport and a decrease in transport expenditure. As anticipated, the increased use of public transport is directly linked to the increased budget facilitated by the travel vouchers. Nevertheless, a crucial factor propelling the increase in ridership is the heightened satisfaction with the public transport service. Given that the sample comprises frequent public transport users, these variables are intricately associated with this specific system.

Satisfaction emerges as a mediator between subsidies and ridership. On one hand, travel vouchers can increase satisfaction levels, particularly in terms of transport and life satisfaction. Then, changes in satisfaction with life and transport can trigger behavioral changes, leading to increased public transport ridership. However, it is imperative not to overlook the significance of service quality, as it may exert a positive influence on future use, surpassing the impact of satisfaction (de Ona, 2021). In essence, passenger satisfaction is not guaranteed if the quality of the public transport service fails to enhance passenger perceptions of the value relative to the fares paid. An affordable fare emerges as a key indicator for improving system satisfaction and driving ridership. Moreover, it may contribute to a positive effect on fare evasion, as previous studies have identified the perceived value of the fare as a determinant of evasion probability (Guzman, Arellana, & Camargo, 2021). Travel vouchers not only increased system usage due to affordability but also enhanced satisfaction with transport and life. This means that if the policies are accompanied by measures to improve the public transport service to meet the needs of travelers, the more the public transport system is used, the more satisfaction increases, creating a virtuous circle.

Finally, the study is not without limitations, necessitating attention in future research endeavors. The RCT took place in 2021, a period when Colombia was emerging from COVID-19-related mobility restrictions. Although no specific restrictions impacted the study, the economic and social context during this time may have influenced the results. Additionally, the experiment's duration was limited to 4 months, and the effects of travel vouchers may manifest over a more extended period. Furthermore, our experiment only considered two levels of travel voucher values, hindering an assessment of the differential effects of subsidized values. Subsequent research should encompass a broader range of subsidy values to evaluate and identify optimal voucher values. Lastly, in the proposed formulation, the latent variables used to assess satisfaction are measured by a finite set of indicators that might not comprise the overall dimensions of the satisfaction levels. Hence, future research could also focus on increasing the number and diversity of indicators and concepts under evaluation and also focus on long-term effects.

Author statement

Luis A. Guzman: Conceptualization, manuscript writing, analysis of results, content planning and review (corresponding author). Victor Cantillo-García: resources, modeling, validation and review.

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Declaration of competing interest

None.

Data availability

Data will be made available on request.

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